

claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_774b13f4-334\agent-ad705115cb02fa0fe.jsonl`  
- SHA-256: `6b82f89aa88ef94827c9c60916248b577ffd8aba50bdb9cd94e835fb79f1c4f`  
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- project: `wf_774b13f4-334`  
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```

Transcript

1. user (2026-06-12T14:36:57.582Z)

Repo: C:\Users\avidu\Projects\badminton-highlight-indexer (branch docs/pmf-field-signals).

Review these files (read them in full):

- docs/PMF_FIELD_SIGNALS.md (NEW - hypotheses H1-H7, GREEN/AMBER/RED scoring anchors, pre-registered decision rules PROCEED/PIVOT A-C/FALSIFIED, rollup mechanics, guardrails)
- docs/FIELD_VISIT_ANSWER_SHEET.md (NEW - the per-visit capture form a non-technical field associate fills by hand)
- docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md (EXISTING, lightly edited - the recruit hand-out with the 7-question script and ground rules)

Grounding docs (also read): docs/PMF_MARKET_RESEARCH.md (esp. SS6 risks and pricing anchors), docs/NEXT_STEPS.md item 5, docs/COMMERCIALIZATION.md C13 row.

Context: the owner ratified that PMF validation happens by physically visiting Bangalore badminton academies (10-15 visits by a paid non-technical field associate) and reading signals. The kit's job: make 15 visits produce a pre-registered verdict (proceed/pivot/falsified) rather than motivated reasoning. Product = KhelSutra (rally indexing + highlights from static-camera footage; wedge = messy multi-court BYO footage; serving COGS ~\$0.30-0.60/hr via Gemini).

A critic claims the following flaw in the field-validation kit. Adversarially verify by reading the actual files - try to REFUTE it. Default to refuted if the flaw is pedantic, out of scope for a 10-15-visit directional read, or already mitigated elsewhere in the kit.

CLAIM [medium] The associate is paid more when H7 = yes ? incentive sits on the falsification-blocking signal

The flyer tells the recruit up front: "If a conversation goes well, there's a Phase-2 follow-up (coordinating a paid video-recording session) with additional compensation." Phase-2 work exists only for academies that say yes to Q7 ? so the associate has a direct financial interest in producing H7-GREENs. H7 is simultaneously (a) double-weighted as a "behavioral" signal (§2), (b) a PROCEED leg (§3), and (c) the sole blocker of FALSIFIED (Ø H7-GREEN). The kit carefully debiases the per-visit pay ("paid per completed honest visit, not per positive answer" ? good) but reintroduces exactly the same distortion one paragraph earlier on the single most decision-loaded question. A warm-sounding "maybe, come back" nudged to "yes + number" is the cheapest possible upgrade for the associate and moves two verdict rows.

Proposed fix: Decouple: pay any Phase-2 coordination work at a flat rate independent of how many academies opt in, or have the associate record only openness (yes/maybe/no verbatim) while the OWNER makes the actual Phase-2 ask on the follow-up call ? which also matches §4's existing rule that the owner, not the associate, carries the demo.

2. user (2026-06-12T14:37:04.632Z)

docs/pmf-field-signals

M docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md

M docs/NEXT_STEPS.md

?? arial.ttf

?? artifacts/

```
?? docs/FIELD_VISIT_ANSWER_SHEET.md
?? docs/PMF_FIELD_SIGNALS.md
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
```

3. user (2026-06-12T14:37:08.600Z)

```
1      # C13 Field Signals ? what academy visits must tell us (and when to change course)
2
3      **Status:** ACTIVE ? the decision layer for the C13 beachhead validation
4      (owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
5      visiting academies and reading signals on the ground*).
6      **Companions:** [ `FIELD_RESEARCH_ASSOCIATE_FLYER.md` ](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
7      (the recruit hand-out: script + ground rules) .
8      [ `FIELD_VISIT_ANSWER_SHEET.md` ](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
form) .
9      [ `PMF_MARKET_RESEARCH.md` ](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
10
11     **Why this doc exists:** the flyer collects answers; this doc fixes ? **before the
12     first visit** ? what those answers *mean* and what totals trigger which decision.
13     Pre-registering the thresholds is the honesty mechanism: it stops us from reading
14     15 ambiguous conversations as whatever we hoped to hear.
15
16     ---
17
18     ## 1. The hypotheses under test (each maps to script questions)
19
20     | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
21     |---|---|---|---|
22     | H1 | **Recording behavior exists** ? academies already produce fixed-angle / CCTV /
multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
23     | H2 | **Dead-footage pain** ? recorded footage piles up unwatched (the indexing wedge)
| footage is small, or already gets watched | Q2 ? |
24     | H3 | **An audience exists** ? coach, player, or parent would actually consume rally
clips/reels | shrug; "who has time to watch clips" | Q3 |
25     | H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+) leave
a gap we fit | "we use X, it's fine, fully solved" | Q4 |
26     | H5 | **Willingness to pay per match** at a price that clears unit economics | "should
be free" / <?100 everywhere | Q5 ? |
27     | H6 | **Multi-court chaos is a felt problem** (our moat: indexing messy BYO multi-court
footage) | multi-court video already useful to them | Q6 |
28     | H7 | **Behavioral commitment** ? they'll let us record on their court (Phase-2 door) |
polite no across the board | Q7 |
29
30     Unit-economics floor behind H5: serving COGS on the Gemini path is ~$0.30?0.60 per
31     hour of video (? ?25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
32     roughly ?75?150 to process, so **?300/match has real margin, ?100/match is
33     break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.
34
35     ## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
36
37     Score each hypothesis on the answer sheet **from what they said and showed, not
38     what you inferred**. When between two grades, take the lower one.
39
40     | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |
41     |---|---|---|---|
42     | **H1 record** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |
records occasionally (tournaments only) | doesn't record; no plans |
43     | **H2 dead footage** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",
tournament backlog) | some footage, partially watched | no backlog exists |
44     | **H3 audience** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I
```

review with students Mondays") | "could be useful, maybe" | no consumer identified |

45 | **H4 whitespace** | no tool today, or uses one and volunteers a gap we cover (multi-court, BYO footage, price) | uses a competitor, mildly satisfied | competitor solves it, satisfied, sticky |

46 | **H5 WTP** | volunteers ??300/match unprompted, or proposes a budget framing of their own (per-student/per-month) | lands at ~?100 with hesitation | refuses to name a number / "free" |

47 | **H6 multi-court** | complains about it in their own words before/without prompting | agrees when asked | multi-court footage already works for them |

48 | **H7 Phase-2** | yes + gives a name/number/slot | "maybe, come back" | no |

49

50 **Weighting rule (behavior > words):** H7-GREEN and H2-GREEN are *behavioral* signals and count double in your gut-read; H3/H5 answers are *talk* until a Phase-2 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a polite conversation, not a customer.

54

55 **Disconfirming-evidence quota:** every visit sheet must record at least one verbatim *negative* quote (the thing they liked least / wouldn't pay for). If a sheet has only positives, the visit gets re-done or discounted ? per the flyer's ground rule 4, a reasoned "no" is worth more than five vague yeses.

59

60 **## 3. Pre-registered decision rules (n = 10?15 academies)**

61

62 Counts, not percentages ? at this sample size we're reading direction, not estimating proportions. Evaluate after **every 5 completed sheets**; final call at the agreed n. "GREEN" below means GREEN on that hypothesis row.

65

66 | Verdict | Trigger (pre-registered) | What we then do |

67 |---|---|---|

68 | **PROCEED** ? beachhead validated** | ?6 H2-GREEN **and** ?4 H5 at ?300+ **and** ?3 H7-GREEN | Run the camera pilot at the 2?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the single warmest |

69 | **PIVOT A** ? "index the backlog" ? "record + index" service** | H2 mostly RED (<3 GREEN) but H3+H5 healthy on *fresh* recordings | The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo |

70 | **PIVOT B** ? partner/API motion** | H4 RED at ?5 academies (incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship (the PB Vision x PodPlay template from the 2026-06-10 audit) |

71 | **PIVOT C** ? payer is the parent, not the academy** | H5 RED from owners but H3 repeatedly names parents/players asking for clips | Re-test WTP as B2C per-student reels (?/reel to parents) with the academy as channel, not customer |

72 | **FALSIFIED** ? beachhead assumption fails** | <3 academies record at all (H1) **and** 0 H7-GREEN | The "academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further |

73

74 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide explicitly with the owner ? but **the tallies above get computed first**, so the discussion starts from the pre-registered read, not from anecdotes.

77

78 **Competitor-intel side-channel (always-on):** any visit where a competitor is named ? fill the answer sheet's competitor block (what it does, what it misses, what they pay). ?3 consistent reports about one product = update

80 ``PMF_MARKET_RESEARCH.md` §4` with field-verified rows (today's table is web-sourced).

82

83 **## 4. Mechanics**

84

85 - One [``FIELD_VISIT_ANSWER_SHEET.md``](FIELD_VISIT_ANSWER_SHEET.md) per academy, same day as the visit, in the associate's words + verbatim quotes.

87 - The owner (or Claude session) transcribes each sheet into one row of the rollout table below (keep it in this doc or a sheet next to it); tallies and the decision-rule read-out update per 5 visits.

89

90 - Demo asset if a conversation earns it: the real compiled reel

```

91      (`output/MahadevpuraSingles_highlights.mp4`) on a phone ? *show, never promise*
92      (flyer ground rule 3 still applies; the associate doesn't carry the demo,
93      the owner does on follow-ups).
94
95      ### Rollup table (one row per completed visit)
96
97      | # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4 | H5 (?) | H6 | H7 |
Gut | Phase-2? |
98      |---|---|---|---|---|---|---|---|---|---|---|---|
99      | 1 | | | | | | | | | | | |
100
101      ### Tally ? decision read-out (recompute per 5 visits)
102
103      | Metric | Count | Threshold in play |
104      |---|---|---|
105      | H2 GREEN (dead footage) | /n | ?6 ? PROCEED leg; <3 ? PIVOT A leg |
106      | H5 at ?300+ | /n | ?4 ? PROCEED leg |
107      | H7 GREEN (Phase-2 yes + contact) | /n | ?3 ? PROCEED leg; 0 ? FALSIFIED leg |
108      | H4 RED (incumbent satisfied) | /n | ?5 ? PIVOT B |
109      | H1 any recording | /n | <3 ? FALSIFIED leg |
110      | H5 RED + H3 names parents | /n | recurring ? PIVOT C |
111
112      ## 5. Guardrails that bind the field work
113
114      - **Phase 1 records nothing** ? notes only (flyer rule 7).
115      - **Phase 2 consent rules are hard:** written consent from the academy AND adult
116      players; **sessions with minors are never recorded for our library** (training-pool
117      guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
118      - Answer sheets contain personal contact details ? they stay between the associate
119      and the owner; transcribe to the rollup without phone numbers.
120

```

4. user (2026-06-12T14:37:09.849Z)

```

1      # Field Visit Answer Sheet ? one per academy (fill the same day)
2
3      > Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
4      > is a *good* sheet. Companion script: `FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
5      > Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` $2.
6
7      ## Visit basics
8
9      | | |
10     |---|---|
11     | Date / time | |
12     | Academy name | |
13     | Area (neighbourhood) | |
14     | # courts in the hall | |
15     | Spoke to (name + role) | |
16     | Contact (phone/WhatsApp) ? *only if offered* | |
17     | How long the conversation ran | |
18
19     ## The seven questions ? their answers (verbatim where possible)
20
21     **Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?**
22
23     >
24
25     **Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?**
26
27     >
28
29     **Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?**
30
31     >

```

```

32
33  **Q4 ? What do they use today (Machaxi / VISIST / Game Theory / BadPro+ / none)?
34  What's good, what's missing, what does it cost them?**
35
36  >
37
38  **Q5 ? ? Would they pay per match for rally cuts + reel + report? What number did
39  THEY say first (?)? What was their face doing?**
40
41  >
42
43  **Q6 ? Multiple courts in frame ? does that make video useless today? Their words:**
44
45  >
46
47  **Q7 ? Open to a paid recording session on their court? (yes / maybe / no ? and who
48  to call)**
49
50  >
51
52  ## Required: the most negative thing they said (verbatim)
53
54  >
55
56  ## Best quotes (anything vivid, in their words)
57
58  >
59
60  ## Scores (circle one per row ? anchors in PMF_FIELD_SIGNALS.md §2; when torn, score
61  lower)
62  | Hypothesis | Score |
63  |---|---|
64  | H1 recording behavior | GREEN / AMBER / RED |
65  | H2 dead footage | GREEN / AMBER / RED |
66  | H3 audience | GREEN / AMBER / RED |
67  | H4 whitespace | GREEN / AMBER / RED |
68  | H5 willingness to pay ? their ? number: _____ | GREEN / AMBER / RED |
69  | H6 multi-court pain | GREEN / AMBER / RED |
70  | H7 Phase-2 door | GREEN / AMBER / RED |
71
72  ## Competitor block (only if they named one)
73
74  | | |
75  |---|---|
76  | Product | |
77  | What it does well (their words) | |
78  | What it misses (their words) | |
79  | What they pay | |
80
81  ## Your gut read (circle): COLD . POLITE . CURIOUS . EAGER
82
83  ## Phase-2 candidate? (circle): YES ? call: _____ . MAYBE . NO
84
85  ## Anything else worth knowing (layout, lighting, camera positions you noticed, who
86  really decides)
87  >
88

```

5. user (2026-06-12T14:37:10.025Z)

```

1  # Temporary Engagement: Field Research Associate (Sports-Tech, Bengaluru)
2
3  **Duration:** ~2?3 weeks, part-time (flexible hours, mostly afternoons/evenings when

```

academies are active)

4 **Location:** Bengaluru ? in-person visits to badminton academies

5 **Compensation:** ? ___ per completed academy visit + travel reimbursement *(discussed individually)*

6 **Reports to:** Avi (phone/WhatsApp: ___)

7

8 ---

9

10 ## The gig in one paragraph

11

12 We're building an AI tool that takes ordinary practice/match video from a badminton academy and automatically finds

13 every rally, builds highlight reels, and generates match reports. Before building further, we need honest, on-the-ground

14 answers from real academies about how they record, what they'd use, and what they'd pay.

15 **Your job: visit 10?15 good

16 academies, have a friendly 15?20 minute conversation with the owner/head coach using the script below, and write down

17 what they actually say.** No selling. No tech knowledge needed. If a conversation goes well, there's a Phase-2 follow-up

18 (coordinating a paid video-recording session) with additional compensation.

19 ---

20

21 ## Phase 1 ? The conversation (what you actually do)

22

23 **Who to talk to:** the academy owner or head coach ? the person who decides what the academy spends money on.

24 If you get a junior coach, be friendly, gather what you can, and ask when the owner is available.

25

26 **Opening (keep it this honest):**

27 > "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton academies ? it takes normal practice

28 > video and automatically cuts it into rallies and highlights. He asked me to talk to a few serious academies to

29 > understand how you work with video today. Could I ask you a few questions? Takes 15 minutes, and we're not selling

30 > anything today."

31

32 **The questions (ask in this order; the starred ones matter most):**

33

34 1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired person?

35 2. ? **Do you have recorded footage sitting around that nobody has time to watch?** (multi-court hall recordings,

36 tournament days, CCTV) ? roughly how much?

37 3. Who would watch rally clips or highlights if they existed ? coaches for training? players? parents?

38 4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory, BadPro+, just YouTube/phone.)

39 What do you like / what's missing in it?

40 5. ? **If you could hand over a full session's video and get back every rally cut out + a highlight reel + a simple

41 report ? would you pay per match for that? What feels fair ? ?100, ?300, ?500 per match?** (Let THEM say a number

42 first. Note their face, not just the words.)

43 6. Your hall has multiple courts running at once, right? Does that make video less useful today?

44 7. Would you let us record one session here (we pay a small fee for the time) to show you what the tool produces on

45 YOUR court? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

46

47 **After each visit, fill ONE answer sheet ? the template is

[`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md)

```

48      (print one per visit):** academy name, location, # courts, person spoken to + role +
contact, answers 1?7 (their
49      words, not yours), the most negative thing they said (required), exact quotes, your gut
read, and whether they're a
50      Phase-2 candidate. *(How the answers get scored and what totals trigger which decision
lives in
51      [`PMF_FIELD_SIGNALS.md`])(PMF_FIELD_SIGNALS.md) ? that part is ours, not yours; just
capture honestly.)*
52
53      ---
54
55      ## Ground rules & gotchas (read twice ? this is the difference between useful and
useless)
56
57      1. **You are researching, not selling.** If they ask "how much will it cost", say
"that's exactly what he's trying to
58      figure out ? what would feel fair to you?" Never quote a price yourself.
59      2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did
you LAST record a session, and
60      what happened to that file?" gets the truth. Prefer the second kind of question.
61      3. **Don't promise anything.** No features, no dates, no free trials. The only thing you
may offer is the Phase-2 idea
62      (a paid recording session + free sample output) and only as "would you be open to
it".
63      4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a
reason is worth MORE to us
64      than five vague yeses. You're paid per completed honest visit, not per positive
answer.
65      5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.**
Ask what it does well, what it
66      doesn't, and what they pay. Spend extra time here.
67      6. **Multi-court is our favorite topic.** If they complain that video is useless because
"all the courts are in frame
68      and it's chaos" ? capture that complaint in their exact words. That's the problem we
solve.
69      7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ?
notes only. Don't photograph
70      the academy or players unless the owner offers.
71      8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back
to record.
72      9. **Confidentiality:** don't describe the tech beyond the opening line, don't name
other academies' answers, and
73      don't share the answer sheets with anyone except Avi.
74
75      ---
76
77      ## Phase 2 (only for academies that said yes ? separate briefing & pay)
78
79      You coordinate a recording session: we pay the academy a small fee, set up a
phone/camera on a tripod (we provide a
80      one-page capture checklist: behind the court, elevated, whole court in frame, landscape,
1080p/60fps, don't move it),
81      record 1?2 sessions, and later return a sample highlight reel + report to the academy
for free. **Hard rules:**
82      written consent from the academy AND the adult players being recorded; **sessions with
minors are not recorded for
83      our library** ? adult sessions only; the academy gets told plainly what the footage is
for (building and demonstrating
84      the tool).
85
86      ---
87
88      ## Interested?
89
90      Message Avi at ____ with "FIELD RESEARCH" and your availability. First 2 visits are

```

paired (you + Avi or a phone
91 walkthrough) so you're never improvising.
92

6. assistant (2026-06-12T14:37:15.243Z)

Now the grounding docs.

7. user (2026-06-12T14:37:16.225Z)

```
1      # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports
2
3      > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4      > validate live before relying on them for investment/pricing decisions. This is the
5      > market-research-of-record behind the PMF section of the living
6      > [ `COMMERCIALIZATION.md` ](COMMERCIALIZATION.md) tracker ($6). Product framing:
7      > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8      > self-recorded, **static-camera, full-court amateur** racket-sports video and
9      > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10     > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12     > **Confidence convention.** Figures from named industry reports, federations, or
13     > company pricing pages are tagged with source + year. Inferred items are tagged
14     > **(estimate)** / **(low confidence)**. Emerging-category market sizes vary widely
15     > between vendors; ranges are shown rather than false precision.
16
17     ---
18
19     ## 1. Executive summary
20
21     The product sits at the intersection of three favorable trends: (a) a fast-growing
22     **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
23     depending on vendor); (b) a structural shift to **amateur self-recording** on action
24     cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
25     segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
26     106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
27     2024, +45.8% YoY).
28
29     The central tension is the **static-camera constraint**: the tool works best on fixed,
30     full-court amateur footage and *fails* on broadcast/panning footage. This is both a
31     moat (most competitors target team sports / broadcast) and a filter on the addressable
32     user ? the customer must already set a camera on a tripod behind the court. That
33     behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not
34     the median casual player.
35
36     **Recommended beachhead (strongest single):** **badminton coaching academies and
37     serious club/competitive players in India** ? a large, rapidly formalizing academy
38     ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing
39     willingness to pay for coaching, an underserved tooling gap (incumbents are
40     team-sports/broadcast), and a direct-but-immature competitor field. The product is
41     *native badminton-first* ? a genuine edge here. Secondary: **(2) US/global pickleball
42     players & clubs** (capital-rich, fastest-growing, but SwingVision already strong);
43     **(3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).
44
45     **Pricing** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)
46     and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A
47     software-only BYO-camera model undercuts the hardware players; realistic anchors
48     **(estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.**
49
50     **Top risks:** (1) amateurs won't reliably set up a tripod ? TAM collapses to
51     coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality
52     variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia,
53     China) are the most price-sensitive and hardest to bill at Western SaaS prices.
54
55     ---
```



```

56
57   ## 2. Market sizing & trends
58
59   ### 2.1 Sports analytics / sports-video market
60   - **2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business
61     Insights); other vendors USD 5.6?6.0B. *(2025)*
62   - **Growth:** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI
63     ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%);
64     Future Market Insights ? ~USD 36.2B by 2035 (22.1%). *(reports 2024?2025)*
65   - **Video analytics is the leading segment** within sports analytics (Grand View /
66     Fortune BI) ? the directly relevant sub-market.
67   - **Interpretation (estimate):** the *amateur/club auto-highlight* slice is a small but
68     fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The
69     product targets the long tail the big numbers under-serve.
70
71   ### 2.2 Action-camera / amateur recording adoption
72   - **2024 size:** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). *(2024?2025)*
73   - **Sports is the largest end-use segment** of action cameras (GMIInsights, 2024).
74   - **GoPro ~47% share, DJI ~20%** (electroIQ/industry, 2024); phones + cheap tripods are
75     the de-facto amateur rig.
76   - **Implication:** the capture-side behavior the product depends on is already
77     widespread and growing ? *for the segments that record* ? validating "BYO-camera,
78     software-only."
79
80   ### 2.3 Participation & popularity by sport and geography
81
82   | Sport | Participation signal | Source (year) |
83   |---|---|---|
84   | **Badminton** | 220M+ play regularly; 160+ countries | BWF est. via BadmintonBites
(2023) |
85   | Badminton ? China | ~80M registered, up to ~100M (largest base) | industry
compilations (2023) |
86   | Badminton ? India | 20M+ recreational by 2023; academy boom + Khelo India | industry
compilations (2023) |
87   | Badminton ? Indonesia | National sport; ~1M club players | industry (est.) |
88   | Badminton ? Denmark | 100,000+ registered club members (Europe #1) | industry (2023) |
89   | Badminton ? Japan/Korea | Major school-club + workplace sport | industry (2023) |
90   | **Tennis** | **106M players (2024)**; +25.6% over 5 yrs (87M in 2021) | ITF Global
Tennis Report (2021, 2024) |
91   | Tennis ? Asia | ~1/3 of players (33M of 87M) | ITF (2021) |
92   | **Pickleball ? US** | **19.8M players (2024)**; +45.8% YoY, +311% over 3 yrs; ~6.2M
core | SFIA / Pickleheads / The Dink (2024) |
93   | Pickleball ? US courts | ~70,000 courts, +55% places to play YoY | SFIA / Pickleheads
(2024) |
94   | Pickleball ? global | IFP 78 member countries; UK +73% in 2024; ~282M played monthly
per PPA claim *(low confidence)* | various (2024?2025) |
95   | **Table tennis** | 30M+ players; 227 member associations | ITTF (current) |
96   | **Cricket** | 2.5B+ fans; 500M+ in India (mostly *fans*, not self-recorders) | Cricket
Rise / Nielsen (2024?2025) |
97
98   - **Pickleball market value:** ~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR
99     (Market.us/Custom Market Insights, 2024?2025).
100  - **Badminton equipment market:** ~USD 7.23B global (2023), Asia-Pacific dominant.
101
102  **Geographic read:** badminton density is overwhelmingly Asian + Denmark; tennis is
103  broad/high-income; pickleball is US-dominant, capital-rich, globalizing fast; cricket
104  is huge as *fandom* but its amateur self-recording + static-full-court fit is weak
105  (large field, not a fixed-frame court). For *this product's static-camera fit*,
106  badminton/pickleball/tennis/table-tennis are the core; cricket is a long-tail roadmap
107  item, not a beachhead.
108
109  ---
110
111  ## 3. Customer segment evaluation
112

```

113 | Segment | Market-size signal | Willingness to pay | Accessibility | Static-camera fit
 | B2C/B2B | Verdict |

114 |---|---|---|---|---|---|

115 | ****Individual amateur/enthusiast**** | Largest raw TAM (100s of millions) | ****Low**** ? competes with free; ~USD 5?15/mo ceiling | High via app stores; high CAC/churn | ****Mixed**** ? most won't set a tripod | B2C | Volume play, weak economics; use as funnel |

116 | ****Coaches & coaching academies**** | India academy boom; thousands across Asia | ****High**** ? already sell coaching; USD 300?1,500/yr plausible | Reachable via federations/distributors/social | ****Strong**** ? already film fixed-angle | B2B/B2B2C | ****Best fit**** ? buyer, recorder, payer aligned |

117 | ****Clubs & leagues**** | Denmark 100k+; thousands of clubs Asia/EU | Medium-high; USD 500?3,000/yr | Medium ? committee sales cycle | ****Strong**** | B2B/B2B2C | Strong secondary; longer cycle |

118 | ****Tournament/event organizers**** | Many small/regional events | Medium ? per-event/seasonal | Medium ? episodic | ****Strong**** | B2B | Good per-event upsell; lumpy |

119 | ****Schools & universities**** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | ****National/regional federations**** | Few, high-value | Medium-high; procurement-heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | ****Broadcasters / streamers**** | High value per deal | High | Low ? incumbents entrenched | ****Poor** ? panning = failure mode** | B2B | ****Avoid**** ? violates core constraint |

122

123 ****Key dynamics:**** the static-camera constraint is a ****B2B filter**** ? coaches/academies/
 124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest
 125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters
 126 explicitly violate the constraint ? exclude.

127

128 ---

129

130 **## 4. Competitive & pricing benchmarks**

131

132 | Product | Category | Model | Price | Notes (year) |

133 |---|---|---|---|---|

134 | ****Veo (Cam 3)**** | Team-sports auto-filming | Hardware + sub | Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr | 15,000+ clubs/80 countries; \$115M raised (2025) |

135 | ****Trace**** | Soccer auto-filming | ****Free camera + sub**** | USD 180/yr (Basic) ? 300 (Pro) | Camera leased free with sub (2025) |

136 | ****Pixellot (Air)**** | Auto camera/streaming | Hardware + sub | Camera ~USD 949 + USD 69?167/mo | Institutional/league; 10,000+ clubs (2025) |

137 | ****Spiideo**** | Auto-filming/analytics | Sub incl. camera | ~USD 3,800/yr (1 team) ? 6,100/yr (club) | Institutional budgets (2025) |

138 | ****Hudl**** | Team video analysis | SaaS tiers | From ~USD 400/yr | Broad team-sports incumbent (2026) |

139 | ****SwingVision**** | ****Tennis/pickleball AI, single iPhone**** | Software freemium | ****USD 179.99/yr****; free 2 hrs/mo | Closest analog; 200k+ monthly users; \$10M+ raised (2025) |

140 | ****BadPro**** | ****Badminton AI on phone**** | ****Pay-per-use, no sub**** | Per-analysis (? "< one coaching session") | Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025) |

141 | Niche pickleball/badminton apps | Various | Mixed | Varies | Fragmented; few true auto-highlight tools (2025) |

142

143 ****What buyers actually pay (synthesis):**** consumer software-only anchors at ~USD
 144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,
 145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses *pay-per-use* ?
 146 signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity +
 147 warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and
 148 sells worldwide via SaaS. Realistic anchors ****(estimate):** B2C USD 60?180/yr; B2B
 149 academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline
 150 self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B****
 151 value story more than a B2C one.

152

153 ---

154

155 **## 5. Recommended beachhead**

156

Ranked by ****WTP × participation density × existing recording behavior × static-camera fit × competitive whitespace.****

#1 (strongest) ? Badminton coaching academies & serious players, ****India****
 20M+ recreational players (2023); rapidly ****formalizing academy ecosystem**** + Khelo India tailwind; academies already record fixed-angle; coaches are buyers who monetize instruction (high relative WTP); incumbents ignore badminton; the only direct badminton AI (BadPro+) is early with unproven pay-per-use. Product is ***native badminton-first***.
****Watch-outs:**** price sensitivity (need INR-tuned tiers); sell to the ****academy****, not the individual.

#2 ? Serious pickleball players & clubs, ****US (then UK/Australia/Asia)****
 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts; strong recording/sharing culture; pickleball is on the roadmap. ****Watch-out:****
****SwingVision** already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+ monthly users) ? head-to-head differentiation, not greenfield.

#3 ? Badminton academies/clubs, ****East/SE Asia (China, Indonesia, Malaysia, Japan, Korea) + Denmark****
 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea school-club & workplace systems), mature club infra, fixed-court recording.
****Watch-outs:**** China needs local hosting/partners + is regulatory-complex; SE Asia price-sensitive; Denmark/EU small but high-WTP and easy to pilot.

****Bottom line:**** lead with ****India badminton academies**** (whitespace + behavior + buyer alignment), use ****enthusiast B2C as the funnel****, validate a ****club/academy B2B2C motion****, then expand to ****pickleball-US**** (out-build SwingVision) and ****badminton-Asia/Denmark**** (density).

6. Risks & open questions (what would invalidate the thesis)

- **Tripod/static-camera adoption among amateurs.**** If serious amateurs won't set up a fixed camera, B2C TAM collapses to coaches/clubs. ***Validate:*** observe 3?5 academies + 50?100 enthusiasts on whether they already record fixed-angle.
- **Free / near-free alternatives**** (YouTube, free app tiers, BadPro+ pay-per-use) cap B2C WTP. ***Validate:*** fake-door pricing at USD 60?180/yr.
- **Capture-quality variance = the failure mode**** (gym lighting, off-center angle, occlusion, mixed resolutions). ***Validate:*** benchmark rally-detection P/R on 100+ real amateur clips; define a "good-capture" spec + auto-reject (ties to ``VIDEO_RECORDING_GUIDELINES.md`` + the observability reports).
- **Monetization mismatch in best-participation markets**** (India/Indonesia/China are price-sensitive, hard to bill). ***Validate:*** local pricing/payments pilot; per-academy INR licensing.
- **Competitive response**** (SwingVision or a team-sports incumbent adds racket sports). ***Validate:*** ship a badminton-quality advantage incumbents can't trivially match (offline self-improving detection, per-rally tagging).
- **B2B sales-cycle drag**** for clubs/federations/schools. ***Validate:*** single-coach/academy self-serve before institutional deals.
- **Clean-licensing positioning**** matters mainly to B2B buyers ? confirm it's a real purchasing criterion via 5?10 buyer interviews.

****Cheapest next steps:**** (a) 10?15 discovery interviews with Indian badminton academy owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).

7. Sources

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 241 - Competitors & pricing: <https://www.véo.com/en-us/pricing> ; <https://traceup.com/pricing>
 ;
 242 <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
<https://swing.vision/subscribe> ;
 243 <https://www.badproplusapp.com/>
 244

8. user (2026-06-12T14:37:16.405Z)

```

1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha
track opened**
4      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
5      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**
**PR-2
6      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
7      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
8      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
9      `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
10     **Authoritative roadmap:**
11     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
12     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
13
14     ## Where we are (one paragraph)
15     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
16     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);

```

```

17     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
18     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
19     sign test 5/6 wins  $p=0.109$  ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
20     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
21     battery is done** (see table) ? the moat is quantified.
22
23     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
24     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
25     |---|---|---|
26     | Heuristic (velocity windowing) | 0.657 | 0.157 |
27     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
28     | Two-stage WASB+Gemini refine | 0.83 | ? |
29     | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
30
31     **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
32     wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
33     the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
34     architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
35     only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
36     Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
37
38     ## Prioritized next steps
39     1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
every lever below feeds on it.
40     Priority order for new videos: **a) multi-court badminton** (the target distribution
AND where the ship target
41     lives), **b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
0.909; pickleball labels are
42     corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
43     `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
44     label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
re-run the Gen-0 harness.
45     The paired sign test needs n: at n=10 with 9 wins,  $p>0.011$  ? significance is
reachable this month.
46     2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
47     reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93
48     (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with
49     paired-sign  $p<0.05$  vs heuristic" ? owner to ratify/adjust.
50     3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for
**ActionFormer/ASFormer** (MIT, minutes on the
51     2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing
harness; per-video threshold
52     calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
53     4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
54     badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
55     by C3 (provenance multiplies per sport).
56     5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:

```


41-| C8 | ****Golden-set vendoring spend**** | ****P1**** | ? | ADR-006: ****Roora (Bengaluru)**** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm quality-per-dollar from the pilot; then GS-3 ``HumanVendorProvider``. See ``GOLDEN_SET_VENDORING.md``. |
42-[Omitted long context line]
43-[Omitted long context line]
44-| C6 | ****Inbound HEVC decode**** | ****P2**** | ? ?? | Decoding users' GoPro H.264/****HEVC**** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual; confirm decode-side exposure with counsel before launch. |
45-| C7 | ****Hosting unit-economics**** | ****P2**** | ? | Per-video cost is **cheap** for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | ****Benchmark \$/video & latency on the target VM**** for A/B/C before pricing. See §5. |
46-| C10 | ****Value layer (product surface)**** | ****P2**** | ? | Minimal exports shipped (EDL\JSON\CSV + ``basic_stats``, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |
--
198-
199----
200-
201-## Changelog
202-
203-- ****2026-06-07**** ? Added the market-research-of-record (``PMF_MARKET_RESEARCH.md``):
204: ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
205- "beachhead validation" workstream. Wired the summary into §6.
206-- ****2026-06-07**** ? Created this living tracker. Merged the 2026-05-30 monetization
207- audit + the commercial-readiness review. ****Refreshed since the audit:**** C1 AGPL
208- removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
209- the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
210- ``DEFERRED_HARDENING_PLAN.md``; reflected PR #50/#51 (input-quality reports, value-layer

10. user (2026-06-12T14:39:05.828Z)

Structured output provided successfully

11. assistant (2026-06-12T14:39:19.822Z)

Verdict recorded: the claim is real, severity medium ? the flyer ties the associate's Phase-2 compensation to Q7 yeses while H7 is a PROCEED leg and the FALSIFIED veto, and no existing kit mechanism neutralizes that incentive.